Network Analysis of Information Propagation in the 2024 U.S. Election Twitter/X Dataset

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Abstract

The 2024 U.S. election generated extensive political discourse on Twitter/X, offering a rich dataset for analyzing digital communication patterns. This project applies network science methodologies to the USC X 24 US Election Twitter/X Dataset, which contains millions of tweets spanning May to November 2024. By constructing retweet and reply interaction networks within a defined temporal window, the study aims to characterize network structural properties, detect major user communities, and evaluate diffusion dynamics of highly retweeted posts. Specifically, theoretical diffusion trajectories generated using an SI contagion model will be compared with observed retweet patterns to assess whether propagation aligns with standard network driven spreading behavior. The results will provide insight into how network structure and community organization influence information propagation dynamics.

1 Data

The dataset is organized into directories named part_{number}, each containing twenty compressed files of 50,000 tweets, producing over one million tweets per part. Each tweet record includes identifiers, timestamps, text content, engagement metrics (reply, retweet, like, quote counts), conversation IDs, language, hashtags, mentions, links, media content, and user metadata. Sponsored advertisement tweets are included but contain only textual content. The dataset spans from 1 May to 30 November 2024, allowing high resolution analysis of communication networks across major electoral events.

The dataset enables the reconstruction of retweet, reply, and mention networks. For this project, the focus will be on the retweet and reply network due to its suitability for modeling information flow and propagation dynamics.

2 Objective

This study aims to investigate the structural organization and diffusion dynamics of election related interactions on Twitter/X. The specific objectives are:

Quantify structural properties of the interaction network within a selected temporal window.

Detect and analyze community structure and clustering patterns among users.

Evaluate the propagation dynamics of the most retweeted posts by comparing empirical retweet patterns with baseline SI model predictions.

3 Problem Statement

Twitter/X represents a large scale, dynamic interaction network where users exchange, amplify, and respond to information. During the 2024 U.S. Presidential Election, this network exhibited complex structural patterns that likely influenced how information flowed and propagated. This study uses the USC X 24 dataset to construct a retweet and reply network within a selected temporal window and analyzes the propagation dynamics of the most retweeted posts. Network structural properties, including degree distributions, clustering coefficients, and centrality measures, will be quantified to characterize topology. Community detection algorithms will identify modular structures and their role in shaping information flow. Finally, SI model simulations on the empirical network will generate theoretical diffusion trajectories, which will be compared with observed retweet time series to determine whether viral posts follow standard network propagation or exhibit structural effects on spreading dynamics.

4 Methodology

A single, temporally coherent window (such as one week surrounding a major electoral milestone) will be selected to construct a stable retweet and reply network. The methodology consists of three components:

Network construction and structural analysis: A directed retweet and reply network will be constructed by connecting users to the authors of the content they retweeted or replied to. Key network metrics, including degree distributions, clustering coefficients, path lengths, network density, assortativity, and centrality measures, will be calculated to characterize the topology and identify influential nodes.

Community detection: Community detection algorithms (e.g., Louvain or Leiden) will be applied to uncover modular structures within the network. Communities will be analyzed for size, cohesion, and internal connectivity. Content patterns within major communities (e.g., frequently used hashtags or commonly retweeted posts) will be examined to identify structural and thematic clustering.

Diffusion analysis using SI based baseline modeling: A subset of the most retweeted posts within the selected time window will be analyzed to assess their propagation dynamics. An SI contagion model will be applied on the empirical retweet and reply network to generate theoretical diffusion trajectories representing expected network driven spreading. The simulated diffusion curves and growth rates will be compared with empirical retweet time series. By examining divergences between expected and observed propagation, the study will determine whether the diffusion of these posts aligns with standard, organically emerging network driven processes, or whether deviations suggest structural effects influencing acceleration or concentration of spread.